Bilal Belli

Software Engineer

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Portfolio : bilalbelli.tech

Summary

IT engineer, passionate about web development and interested in IT solutions engineering. I am open to professional opportunities, on fixed-term or permanent contracts, in person or hybrid, that align with my software engineering skills.

Experience

Fullstack Developer - Perfective Maintenance of a Microservices Platform

Berger-Levrault - 6 months, 2025 - Toulouse, France

- Designed a solution to automate the migration of **REST/JSON APIs** to **REST/Protobuf**.
- Developed tools in **Java** for implementing **gRPC** in the **Spring Boot** backend and **Angular** frontend.
- Applied perfective maintenance to the WeGF microservices platform by migrating its APIs to gRPC.
- Achieved up to 82% faster loading times in the WeGF application, validated through performance testing.
- Authored Confluence documentation: installation and user guides, migration steps, tests plan and results.
- References : anas.shatnawi@berger-levrault.com ; benoit.verhaeghe@berger-levrault.com.

R&D Engineer - Reverse Engineering & Cybersecurity

LIRMM & Pradeo - 9 months, 2024 - Montpellier, France

- Developed a tool in **Python** for **detecting** malicious behaviors in iOS applications.
- Designed a **decompiler** to reverse engineer iOS application Mach-O binary files.
- Automated algorithms for inter and intra procedural **static analysis** (Taint analysis).
- Conducted unit, integration, functional, and technical tests, achieving good software reliability.
- Wrote a scientific paper updating the state of the art in malware detection techniques.
- References : abdelhak.seriai@lirmm.fr ; antoine.floch@pradeo.com.

Web Developer - HR Management System

AlgeriaPartners - 3 months, 2022 - Algiers, Algeria

- Delivered a web application (NodeJs, ExpressJs, Bootstrap) to manage company human resources.
- Designed the web application prototype and **UI/UX** interfaces in **Figma**.
- Designed the database schema, implemented functional requirements and deployed the app on Google Cloud.

Education

Master's Degree in Computer Science, Software Engineering — Montpellier, France
University of Montpellier — 2024 – 2025

Engineering Degree in Computer Science, Computer Systems Engineering — Algiers, Algeria
Higher National School of Computer Science (ESI ex INI) — 2019 – 2024

Technical Skills

- Python, Java, C, JavaScript, HTML, CSS, Node.js, Spring Boot, Angular, Bootstrap, Flask
- Database administration : SQL Server, PostgreSQL, MySQL, NoSQL (MongoDB, Firebase, XML)
- DevOps CI/CD : Git, Jira, Jenkins, Docker, Kubernities, Ansible, Maven, Gradle
- Software testing : Postman, DevTools, Pytest, JUnit, Playwright, SonarQube
- Environments : Linux (shell, Ubuntu, Debian, Kali), VMware, GNS3, Nginx

Personal projects

QCM Website Generator (link)

- Developed a Java-based project leveraging Model-Driven Engineering (MDE) to automatically generate a QCM website from simplified human input.
- Defined two metamodels in Eclipse Ecore and implemented a custom DSL using Xtext.
- Performed model-to-model (M2M) transformations with QVT and model-to-text (M2T) code generation with Acceleo to produce the final HTML and JavaScript output.

NetFilter - Chrome Extension (link)

- Security tool that blocks malicious websites while browsing, based on an updated threat database.

Trendly - Tech Trends Website (link)

- Integrated, filtered, and ranked results from multiple APIs to identify trending tech topics.

MediaPlayer - Web Application (link)

- Tool for digital ad banners that allows displaying media elements with precise timing and order.

Dictionary Wifi Attack Tool (link)

- A proof of concept GUI tool to test a large number of passwords to connect to a WiFi network.

Academic projects

NetBoost - Content Delivery Network Architecture Simulation

- Simulated a CDN using GNS3 and virtual machines for : cache servers, an origin server and a backup server.
- Used Nginx as a reverse proxy, Varnish for caching, Bind9 DNS server and Apache2 as a web server.

Automatic PDF Summary Generator using LLMs

- Developed a web application (Flask, NuxtJS, Ngrok) integrating Gemini and Ollama LLM's to : analyze and discuss document annotations, generating a concise final summary. (link)
- Achieved 73% accuracy compared to manually annotated summaries.

Bin Packing Problem Optimization

- Developed a comprehensive set of solutions and implementations for the 1D Bin Packing problem, including Best Fit, First Fit, Next Fit, Worst Fit, Branch and Bound, Tabu Search, and Genetic Algorithm.
- Proposed a hybrid metaheuristic approach combining World Water Optimization and Simulated Annealing. (link)

Convolutional Neural Networks (CNN)

- Developed and trained two CNN models in Matlab for Diabetic Retinopathy classification : one for 5 severity levels and another for 3 categories. (link)
- Designed and trained a CNN model in Python to recognize arabic alphabet characters. (link)

Extra Activities

Freelance – Upwork (link)

- Troubleshooting and fixing errors and bugs in a Java SaaS application
- Developed a web application for an online store using the Stripe payment API.

Online Course Author - Udemy (link)

- Created a course explaining the principles of Virtual Reality and Augmented Reality.

IT Security Mentor – Shellmates Club

- Introduced IT security to new club members.